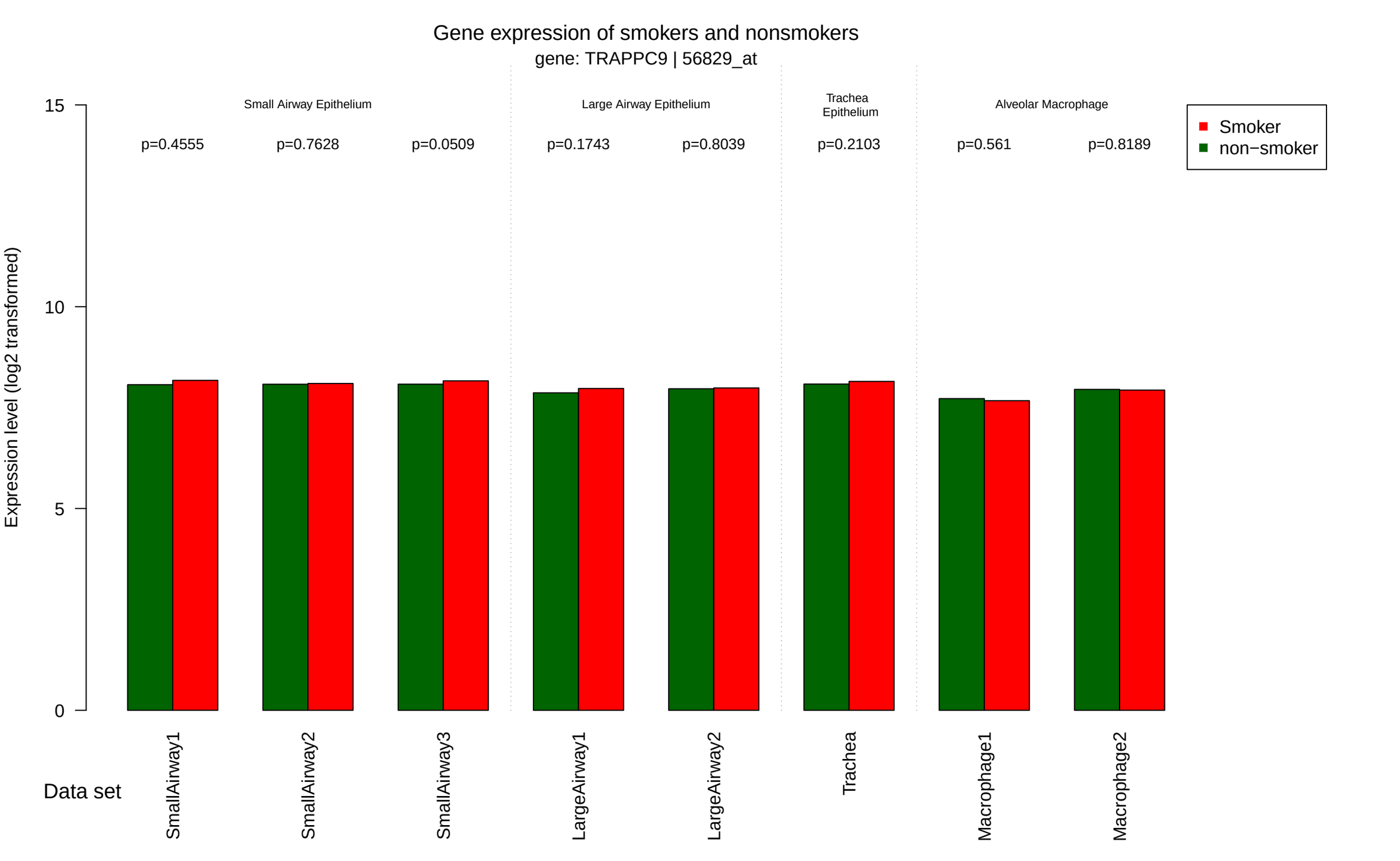
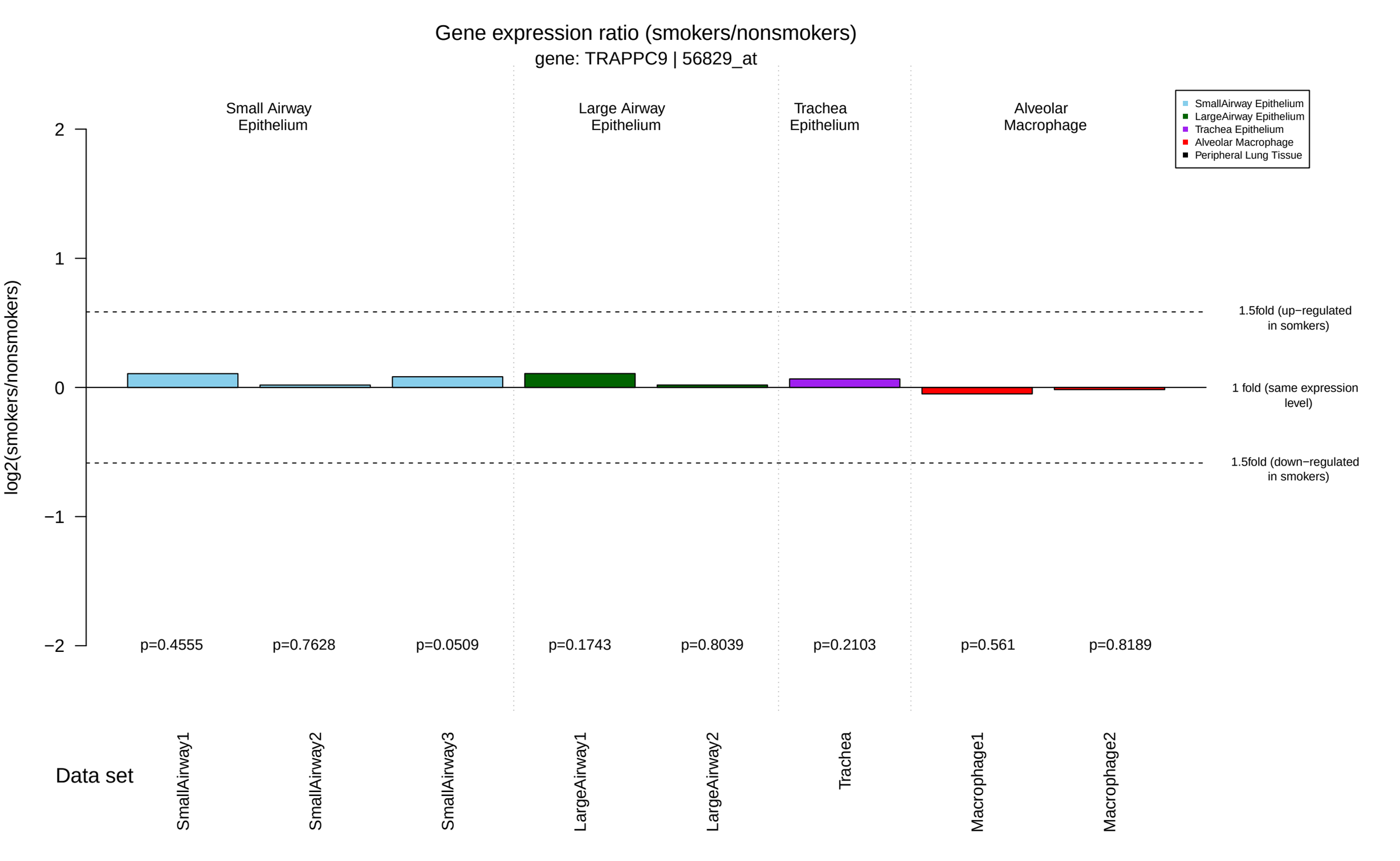
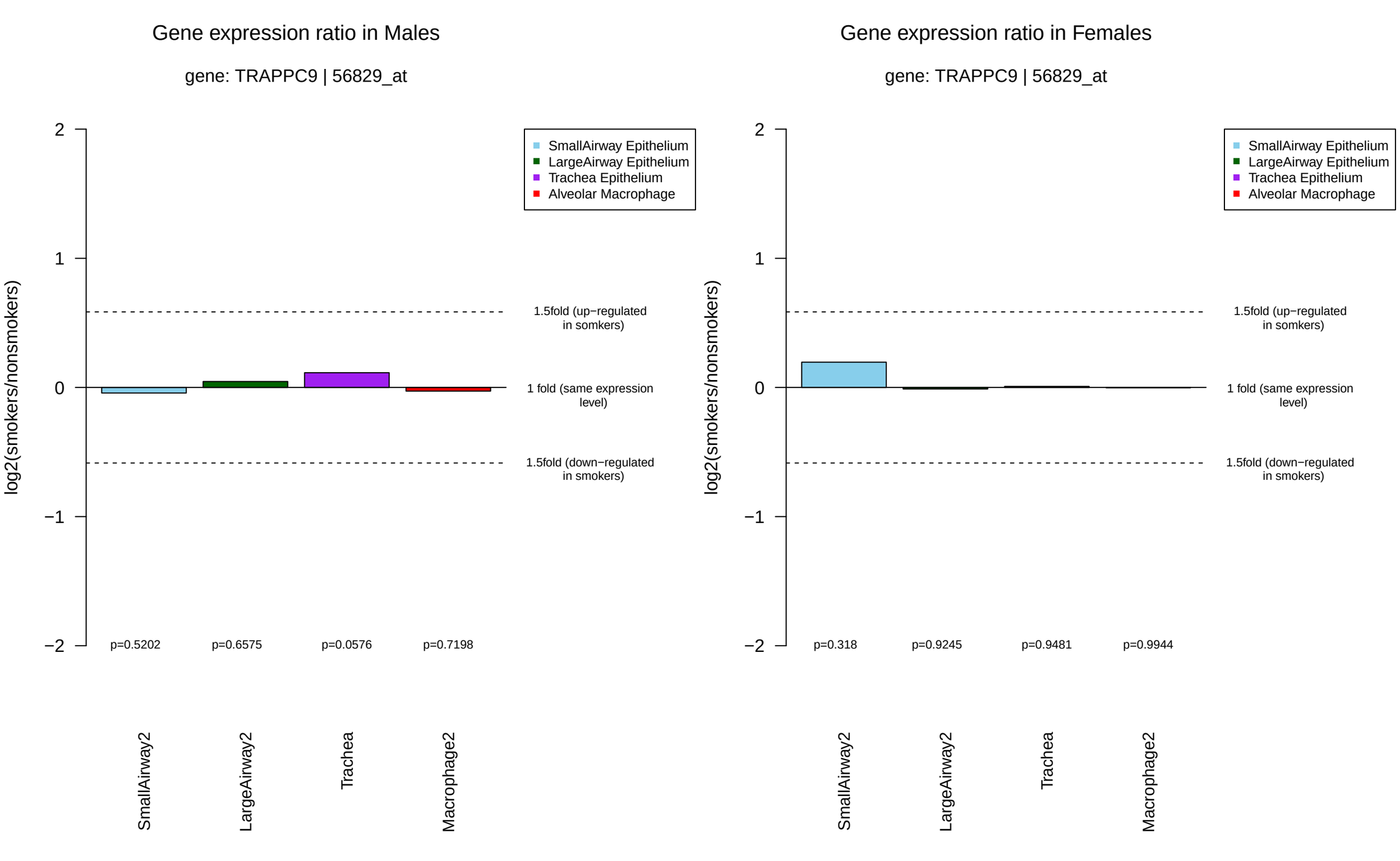
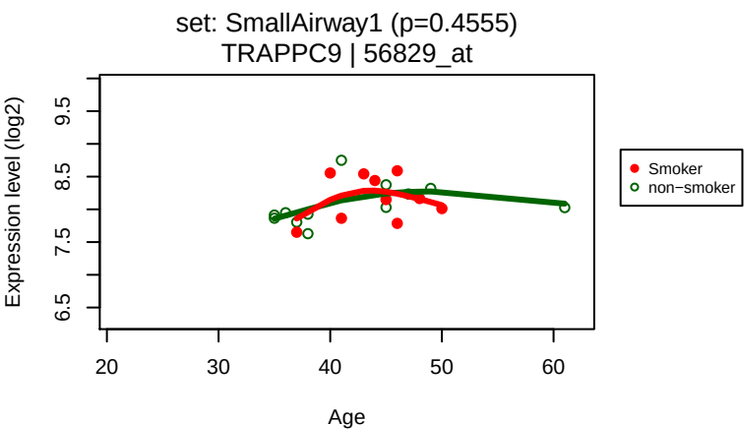
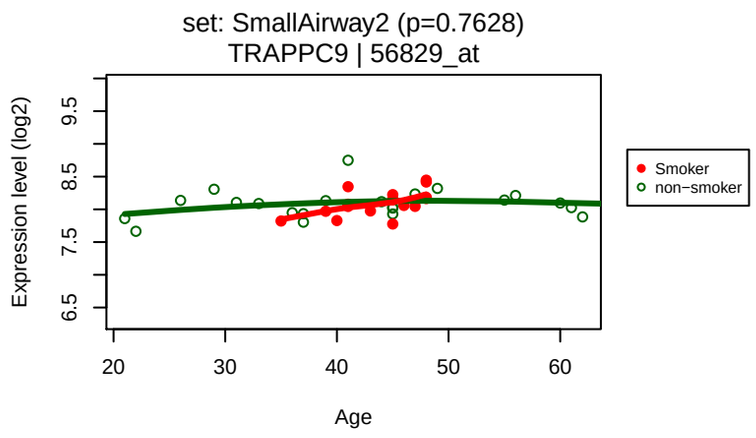
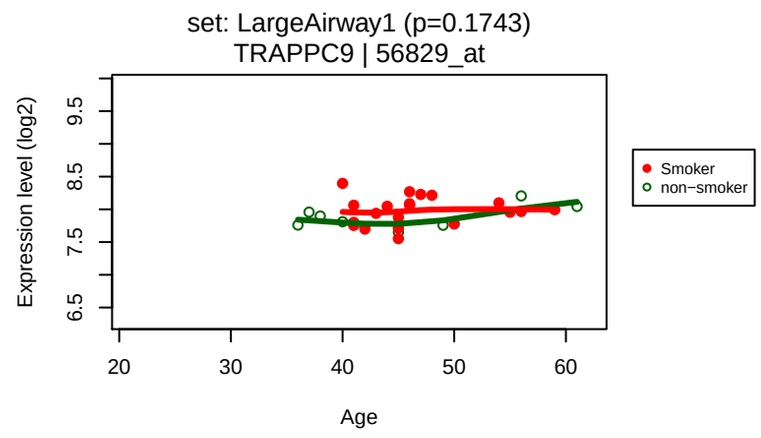

Gene expression of smokers and nonsmokers
gene: TRAPPC9 | 56829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4555
p=0.7628
p=0.0509
p=0.1743
p=0.8039
p=0.2103
p=0.561
p=0.8189
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

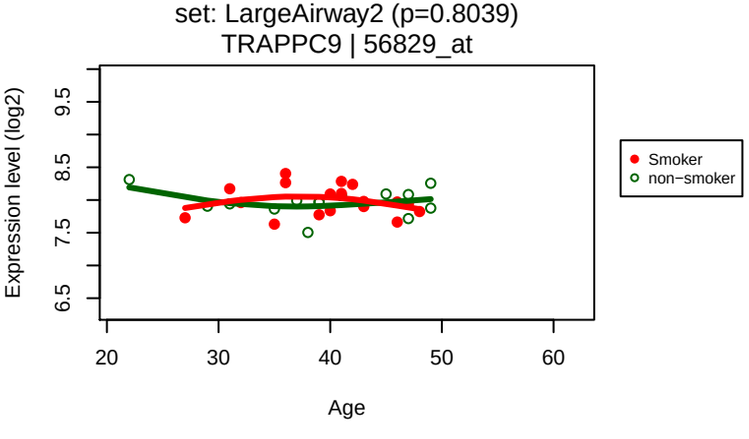

Gene expression ratio (smokers/nonsmokers)
gene: TRAPPC9 | 56829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4555
p=0.7628
p=0.0509
p=0.1743
p=0.8039
p=0.2103
p=0.561
p=0.8189
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

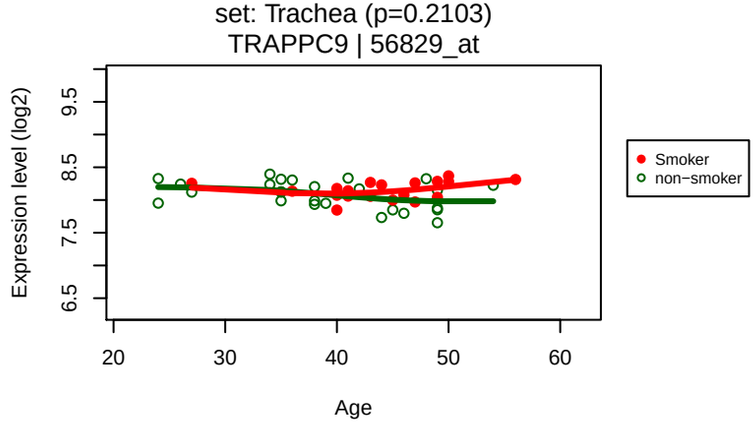

Gene expression ratio in Males
gene: TRAPPC9 | 56829_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5202
p=0.6575
p=0.0576
p=0.7198
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAPPC9 | 56829_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.318
p=0.9245
p=0.9481
p=0.9944
SmallAirway2
LargeAirway2
Trachea
Macrophage2

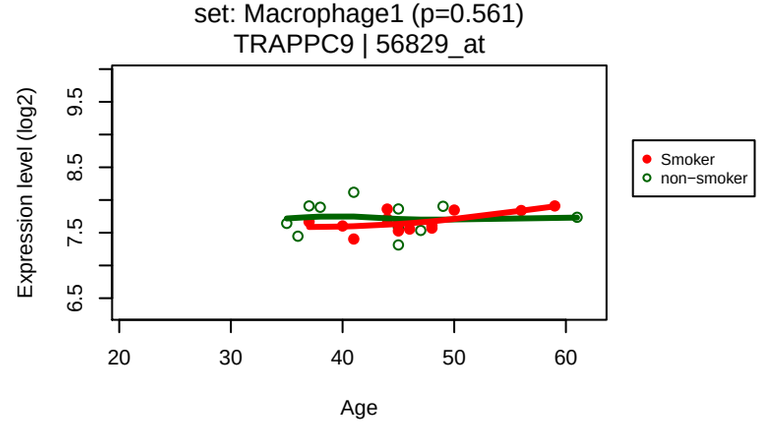

set: SmallAirway1 (p=0.4555)
TRAPPC9 | 56829_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

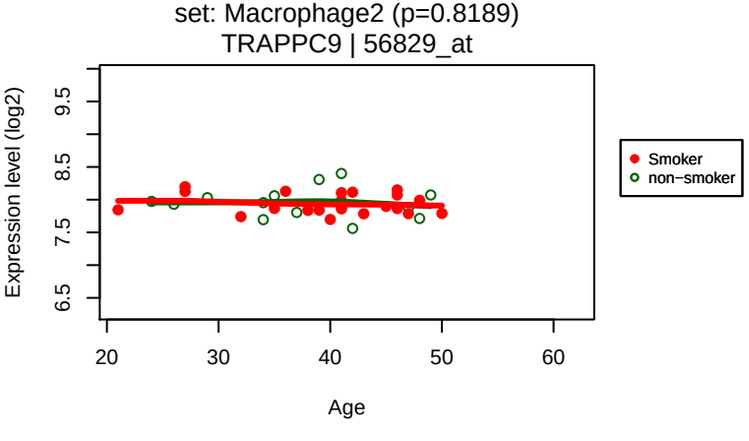

set: SmallAirway2 (p=0.7628)
TRAPPC9 | 56829_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

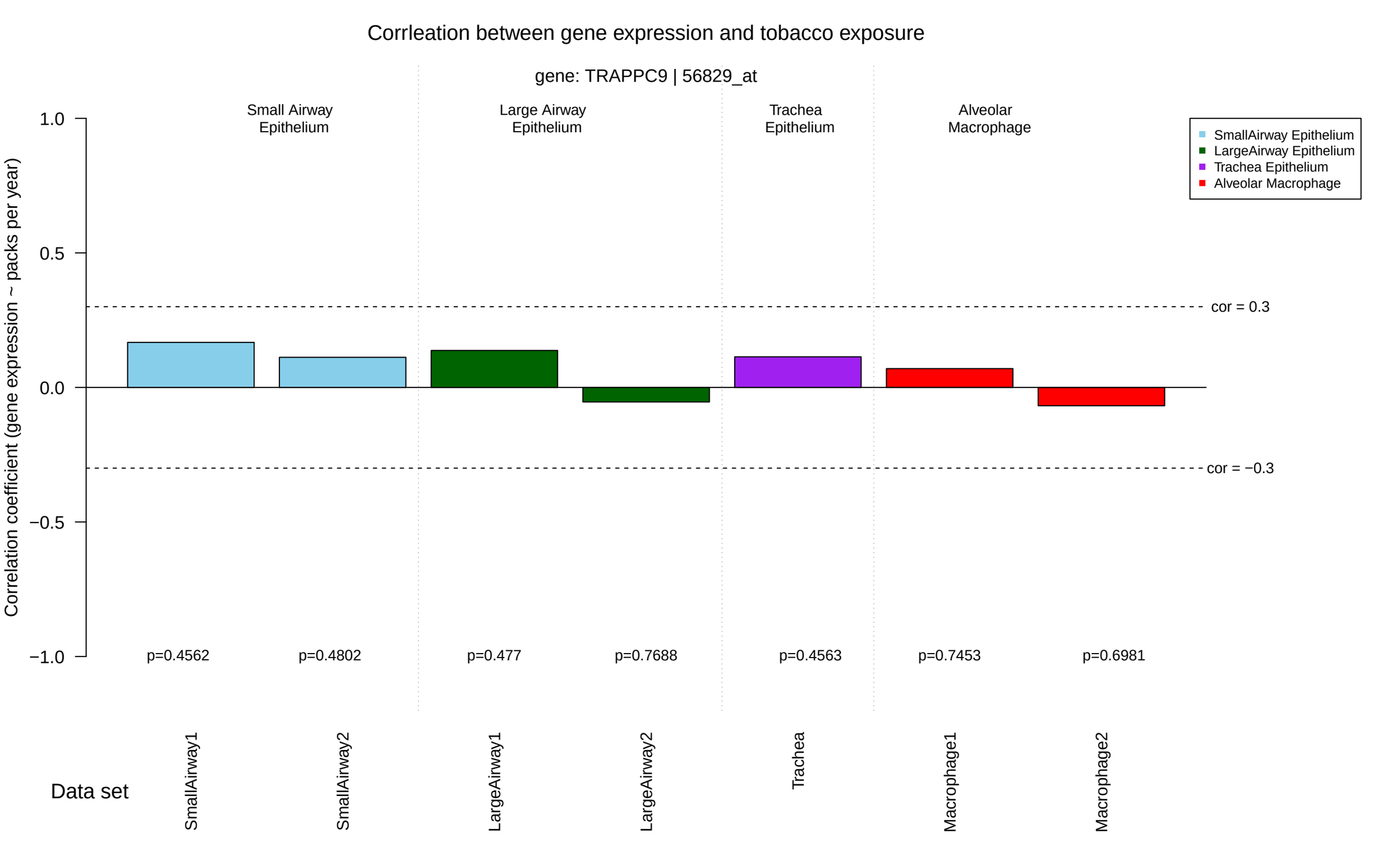

set: LargeAirway1 (p=0.1743)
TRAPPC9 | 56829_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8039)
TRAPPC9 | 56829_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2103)
TRAPPC9 | 56829_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.561)
TRAPPC9 | 56829_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8189)
TRAPPC9 | 56829_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAPPC9 | 56829_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4562
p=0.4802
p=0.477
p=0.7688
p=0.4563
p=0.7453
p=0.6981
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2